IMT 561 Data Visualization: Design And Development Spring 2026

HWK 4: Interactive Visualization

In this homework, you'll design and implement an interactive visualization that shows the current time in a creative, personal, or unconventional way.

By completing this assignment, you will:

- Connect ideas of information representation and perception of time through the design of visualizations.
- Experiment with new ways to represent time, beyond clocks and calendars, by exploring how visual encoding choices (shape, color, motion, and layout) can express abstract data.
- Practice using code and interactivity to make a design feel alive and responsive, strengthening your skills in building dynamic, real-time visualizations.
- Refine your ability to iterate and evaluate design choices based on clarity, meaning, and user experience–core goals of effective information visualization.

Working with clocks is a great exercise for information visualizers and designers because time is both familiar and abstract–everyone understands it, yet there are endless ways to express its passing. It challenges you to think about how data, motion, and meaning can come together in a clear and visually engaging way. This homework also helps you practice interactivity, learning how user input, motion, and timing can work together to make a visualization feel responsive, dynamic, and alive.

Before you begin, you are asked to enrich your concept of clocks and timekeeping by reviewing and/or examining the following resources:

- [ ] Spend a few seconds reflecting on the design of each of the examples in the Appendix at the bottom of this document.
- [ ] Inspect: http://www.thecolourclock.com/
- [ ] Check our class template examples: https://bcsaldias.github.io/imt-561-homeworks
- [ ] Fork class repo: https://github.com/bcsaldias/imt-561-homeworks . To this repo, I have added examples of clocks to scaffold your learning, to play with and inspect. Please carefully check and read those examples. There are three tabs where you'll be implementing your homework. You can merge the changes into your repo or rebase, or fully replace; whatever is easiest for you. **Just make sure that your code is always being deployed to your course github page.**

Before Coding

- **Context:** Think about where your clock belongs, who is the audience? E.g., mountain climbing, hiking, running, morning routine, pomodoro, yoga routine, breathing, boardgame nights, cooking, studying, meditation, gardening, or music practice. Once you've chosen your context you can move to the next step. You will design your clock for a context of your own choosing. As you brainstorm ideas for design, you can think about: Where would it be? On the side of a building? As a piece of furniture? In someone's pocket? As a digital tattoo? As a smartwatch? As a projection on a wall? As a piece of jewelry or wearable art?
- **Sketch** first on paper. Create 10 sketches of different clocks that would work in your context but *in different scenarios or activities within that context*, limit yourself in time (no less than 5 minutes each sketch but also no more than 10 minutes), make annotations on why the sketch serves your context and activity in specific.
    - You'll submit pictures of all your 10 sketches.
    - Try to devise a novel graphic concept. I encourage you to seriously question basic assumptions about how time is represented.
    - Stuck for ideas? It's OK. Riff on the classic analog (circular) clock design. Make an interesting alarm clock.
- [Coding] Choose **TWO** sketches (with some feedback from others) and implement a basic functional version of each.
- From those two, choose **ONE** final prototype, your favorite, explain why it's your favorite, and why it fits your context so well.
- Iterate on your final clock 3-4 times by getting feedback from others (anyone, but it needs to be documented). Think about how the design works with colors, shapes, and your context.

During Coding

- **Create** your clocks in [p5.js](p5.js).
- **Limit** your design to a canvas which is *no larger than 800 x 800* pixels, please. Within this limitation, your clock may have any aspect ratio you please.
- Feel free to experiment with any of the graphical tools at your disposal, including color, shape, transparency, etc. Reactivity to the user/cursor/etc. is optional. Naturally, you'll need to use the `hour()`, `minute()`, `second()`, and `millis()` functions, but you're also welcome to use `day()`, `month()`, and `year()` functions in order to build a clock that evolves over longer timescales.
- Students are encouraged to work with co-pilot in Visual Studio Code, in the mode of ASK, not in the mode of AGENT. Or use any other Gen AI tool for coding. This is a low stakes task and Gen AI is good at following very specific instructions in P5.JS.
- Important note: You'll track your commits, and each commit needs to include **a single feature** in your clock. Anytime that you implement a new piece of code and test it (either from your own coding or with help from the ASK Gen AI helper, you'll commit your code with a clear comment). Your comment needs to indicate what design change you have made. Commits that add more than 3 features at a time will be discounted in the iteration and coding process.

- Check for an example: [commits/main/sketches/sketch13.js]
- Your sketches will be in P5.JS **instance mode**, not global mode. And will be coded within the pages: **sketches/sketch2-hwk4-1.js** and **sketches/sketch3-hwk4-2.js** in the template repo.

Make it interactive (extra credit)

- Bring your clock to life for the users! Use user input like the mouse, keyboard, or buttons to let people interact with your design. Think back to Lecture 7, when I displayed a circle following the mouse (mouseX, mouseY), or to the Pomodoro clock example with clickable buttons. Add a small interactive feature that fits naturally with your clock's context.

Annotations in web portfolio

- In the web portfolio, explain the context in which your clocks / timers would be used.
- For each of the two final clocks:
  - Sketch iteration documentation: Document your work by embedding images of paper/hand-drown sketches corresponding to the specific clock. These could be as simple as photos of your sketches, captured with your phone.
  - Write a paragraph or two (or a series of bulletpoints) explaining your design process for the iterations and final product making explicit the information visualization principles and considerations are present. Add a self-reflection on what you'd like to improve in your final artifact.

Official Deliverable

- [ ] Public repo with two clocks + annotations
- [ ] Deployed website with full documentation (see tutorial for deployment)
- [ ] Slides with your 10 sketches by hand.

Acknowledgement

Thanks to Char Stiles and Ali Klemencic for their brainstorming around this homework, which draws from Ben Fry's and Golan Levin's classes.

Rubric

Total: 30pt

Code History and Deployment (4pt)

- Accessible repository that has three clocks (1pt)
- Must include repo and deployed page link in slide deck, else we'll discount a point.
- Each commit has a clear description of design choice and changes (1pt *3)

Writeup (26pt)

- Context description 3pt

    - 1pt for complete description
    - 1pt description that specifies personal interest
    - 1pt for clarity
- Sketches
    - 0.5pt * 5 = 2.5pt, **due EOD on Wednesday Apr 29, 2026.**
    - 0.5pt * 5 = 2.5pt
- Final selection of sketches (6 *2 = 12pt)
    - Clear explanation that provides reasonable rationale and takes into account feedback (3pt)
    - Designs are quite different from each other (3pt)
        - Design is unique
        - Any change that is limited to one simple change (e.g., color, position) cannot be counted as a unique design.
- Final clock (6pt)
    - A polished version of the final clock (3pt)
    - A clear explanation on design considerations (3pt)
        - 1 pt for complete
        - 1 pt for clarity
        - 1 pt for convincing description of design rationale

Discounts

While the above gives points for completion and we aim to reward all the great work students do, we'll also grade for thoughtfulness and learning. This means discounts will be applied in specific scenarios; see below.

Process and craft oversights (up to -3 points)

- Missing or incomplete commit history, no evidence of iteration or feedback, unclear or broken code, failure to work in instance mode, repo or deployment not public / accessible.

Design and concept oversights (up to -3 points)

- No clear mapping between data and visuals. E.g., time variables (hour(), minute(), second(), etc.) not meaningfully encoded visually.
- Overly literal or conventional design: simple analog clock reproduction without any conceptual or contextual innovation.
- Lack of contextual grounding: design doesn't reflect or make sense within the chosen context (e.g., yoga, mountain climbing, etc.).
- Minimal variation between versions: iterations only change color or size but not concept, structure, or encoding.
- Visual clutter or confusion: design choices make it hard to perceive or interpret time; violate clarity or perceptual principles.

Reflection and communication oversights (up to -3 points)

- Incomplete or missing written reflection: missing required paragraphs for sketches or final clock.
- Weak design rationale: explanations that fail to connect design decisions to course concepts (representation, perception, encoding).
- No mention of the feedback process: iteration not supported with feedback evidence.
- Unclear or poorly organized slides: slides missing required elements or lacking readable annotation and images.

Conceptual oversights (up to -3 points)

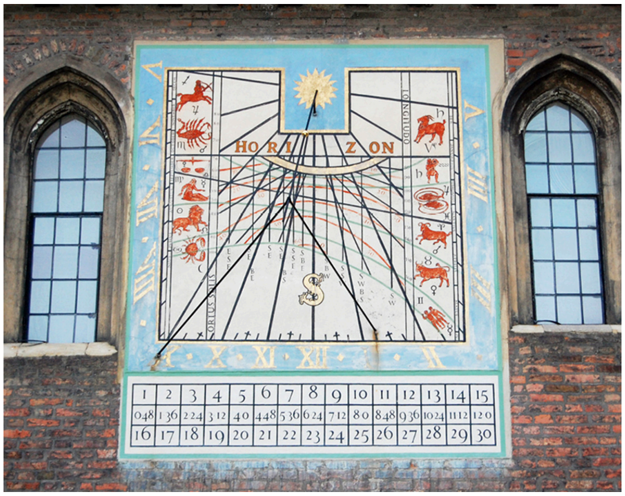
- No sense of experimentation: no evidence of trying different approaches or “questioning assumptions” about time representation.
- No personal or expressive aspect: visualization feels generic, disconnected from the chosen context.
- Fails to feel dynamic or “alive”: lacks motion or interactivity where appropriate, contradicting the goal of designing responsive, temporal visualizations.

Appendix

Smartwatch clock

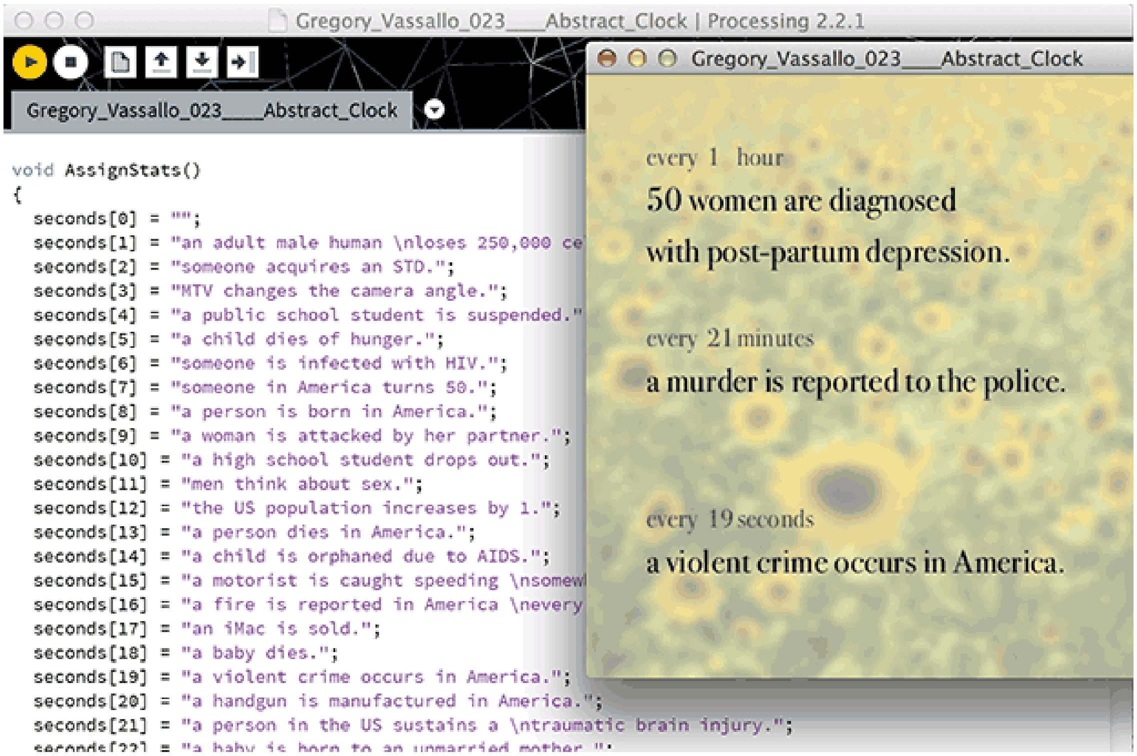
Military history, noonday gun

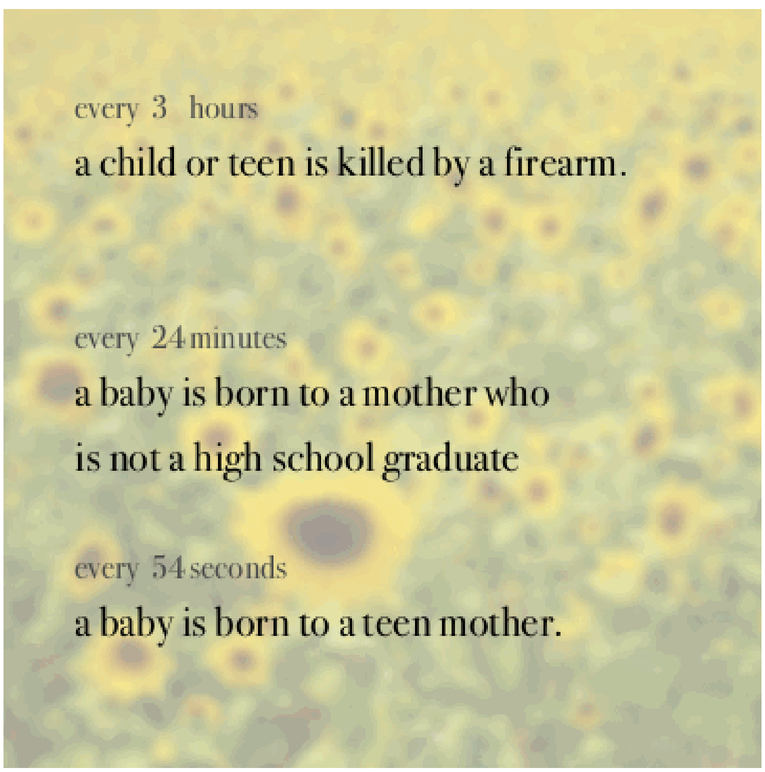
Read shadows by the moon / Opening and closing of flowers



12 is Arbitrary! It’s worth remembering that the convention that we use 12 (or 24) hour time is totally arbitrary — an artifact of ancient Egyptian astronomy. Here two 10-hour clocks.

Examples of clocks, by Greg Vassallo.

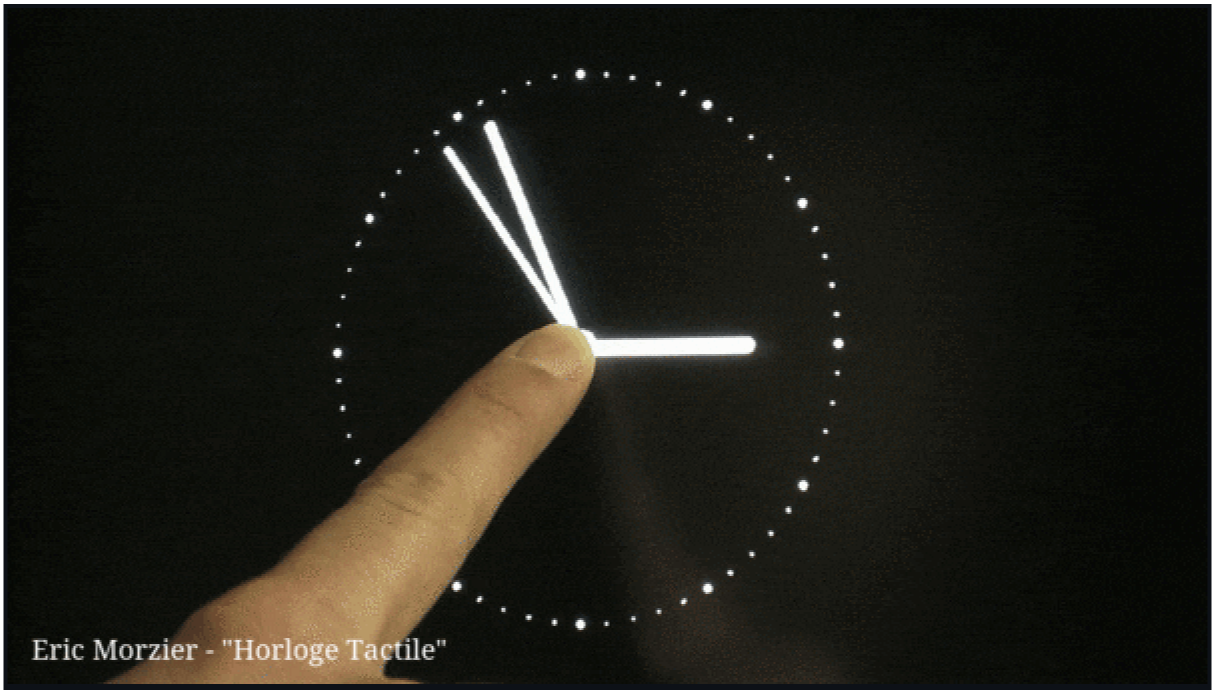




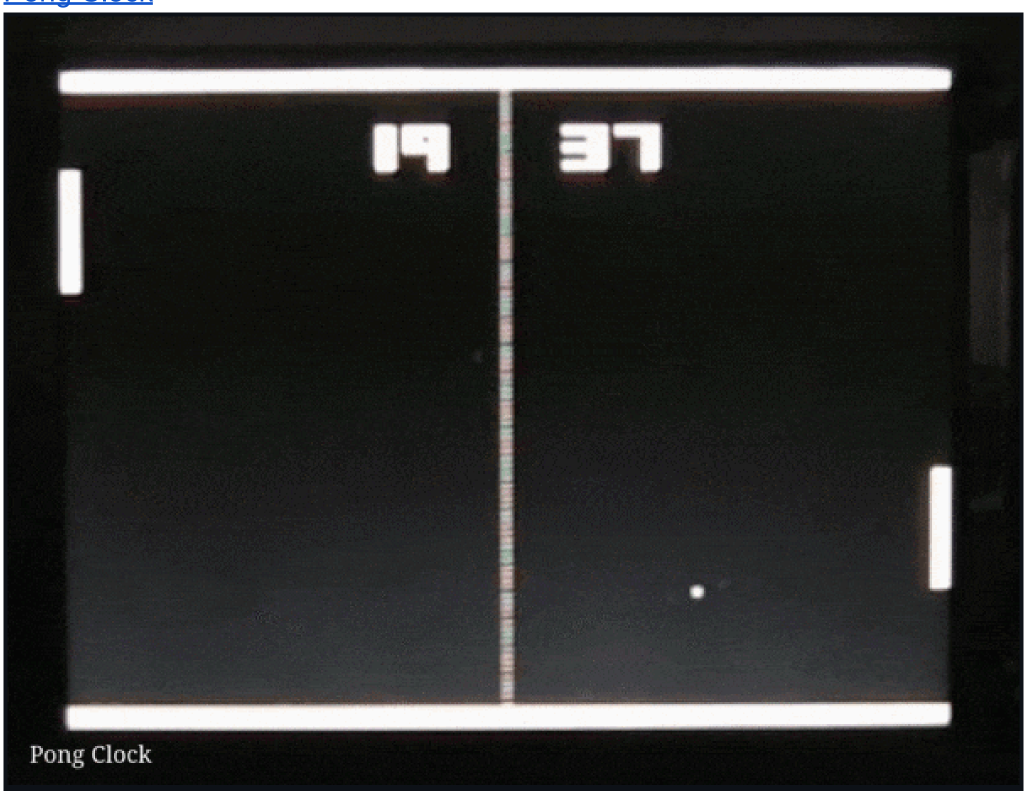
Taiwanese designer Yen-Wen Tseng has designed a clock, Hand in Hand, where the hands are linked by two pivoting arms.

It's about time by insightoutsight / Laurence Willmott

Horloge Tactile by Eric Morzier



Pong Clock



Alicja_Kwade-clock.gif - Against the Run and Gegen den Lauf by Alicja Kwade

A 24-hour clock with a customizable dial to set according to how you spend your day. See more.

Much of this kind of investigation began with John Maeda's highly influential 12 O'Clocks software from 1996. Here's a video (jump to 4'00"), and here are individual GIF recordings: maeda-01, maeda-02, maeda-03, maeda-04, maeda-05, maeda-06, maeda-07, maeda-08, maeda-09, maeda-10, maeda-11, maeda-12